

Teacher's Gradebook and Report Card System Plan

EduJournal

1. Project Overview

1.1 Project Title: EduJournal

1.2 Problem Summary:

In educational systems, teachers enter students' performance or marks manually by writing on paper or computers. And they are calculating average and grades are also manually. That method is not hundred percent accurate and not very efficient, it will make teachers tired and waste a lot of time and, on the other hand, it is difficult with large numbers of students.

1.3 Intended Audience/Users:

- The main users are teachers. They can use this application to enter the marks, edit and calculate average marks and generate the grades and report cards. They can give feedback/ leave comments if they need to.
- Admins can create courses and assign teachers to their role inside of the system and to courses.
- Students are passive in this system, they can access the system to view their marks, grade, average and feedback too. They are not allowed to edit.

1.4 Main Features:

- Grading weight of the exam
- Recording students' grade for assignment, tests and group projects
- Calculating averages and grades
- Generating final grade and report card
- Providing feedback
- Using MariaDB for managing DB and JavaFX for coding

2. Project Objectives

- To design a digital gradebook system
- To automate the calculation of student averages and grades
- To enable teachers to enter and edit marks
- To generate report cards based on students' performance
- To allow teachers to provide written comments
- To allow students to view and download their grades, comments and report cards
- To reduce manual errors and increase efficiency

3. Scope and Deliverables

3.1 Project Scope

This project focuses on developing a Java-based project Teacher's Gradebook and Report Card System. The purpose of the system is to help teachers easily manage student information, record grades, calculate results, generate report cards and students to view the results through a simple desktop application

The application will be built by using JavaFX for the user interface, JPA (ORM) for database connectivity and MariaDB for data storage. The system is mainly for teachers, will support basic academic grading activities and for students to check their grading.

3.2 What the project includes

- A desktop application developed using JAVA FX
- A relational database (MariaDB) to store data of student's
- Teacher, admin and student login functionality
- Teacher usage to add, edit and delete student records and Student can view
- Grade entry update for assignments and exams
- Automatic calculation of student averages and weighted grades
- Report card generation based on student performance
- Basic unit testing using Junit
- Use of Git for version control and Trello for sprint management
- Automated build and test execution through a CI/CD pipeline

3.3 What the project does not include

- Web or mobile application version
- Cloud based deployment

3.4 Project Deliverables

- A requirement and planning document
- Database design
- A working JavaFX application
- Grade Management and report card features
- Unit test cases and test results
- Final project documentation
- A live demonstration and presentation

4. Project Timeline (Trello)

The project will be completed over a period of eight weeks. Following an agile approach with four sprints, each lasting two weeks. We will use Trello to plan tasks, track progress and manage sprint activities.

4.1 Sprint 1: Requirement and Planning (Weeks 1-2)

During sprint one the focus was on understanding the project and planning the work for upcoming sprints.

Completed Milestone:

- Project requirements analyzed and gathered.
- Created User stories and product backlog in Trello
- Defined Project Vision
- Created Project Plan
- Created key UI Design elements (partly)
- Designed DB structure

Dependency:

- Completion of sprint 1 planning is required before starting system design and development in sprint 2

4.2 Sprint 2: Design and Core Development (Weeks 3-4)

During sprint two the focus was on designing the system and implementing the core functionality.

Completed Milestone:

- Database schema is finalized.
- MariaDB tables created.
- JavaFX user interface developed
- JPA connection established between JavaFX and MariaDB

Dependency:

- Sprint 2 development depended on the completion of Sprint 1 requirements and planning.

4.3 Sprint 3: Feature implementation and Testing (Weeks 5-6)

Sprint 3 focused on adding key features and testing the system.

Completed Milestone:

- Student and grade management features implemented
- Grade calculation logic completed.
- Report card generation implemented
- Unit testing performed using JUnit

Dependency:

- Sprint 4 development depended on the completion of database and UI development.

4.4 Sprint 4: Finalization and Presentation (Weeks 7-8)

Sprint 4 focused on finalizing the project and preparing for submission.

Completed Milestone:

- Final features integrated.
- Bugs are fixed and system established.
- Project documentation completed.
- Final project presentation completed.

Dependency:

- Successful completion of all previous sprints.

5. Resource Allocation

Our team consists of 4 members. Our team: Olena, Bayram, Grigorii and Maria. In this project, we follow an Agile approach, and our work is divided into sprints. Each team member has a specific role and works collaboratively. But roles are flexible, and members may contribute to multiple areas depending on sprint needs.

5.1 Team Members and Roles

Scrum Master

Each team member will be in a role as a scrum master. His/her responsibilities are:

- Facilitates Agile planning and sprint activities
- Organizes sprint planning, reviews, and retrospectives using Trello
- Ensures sprint goals align with project objectives and deadlines

Backend Developer

As a backend developer member is responsible for:

- Develops core application logic using Java
- Implements grade calculations, weighted averages, and grading rules
- Integrates report card generation features incrementally
- Collaborates with the database and UI components throughout sprints

Front-End Developer

Developer ensures that team achieves following goals:

- Designs and implements JavaFX user interfaces
- Collaborates with the team to refine UI features during each sprint
- Ensures usability, accessibility, and consistent design

Database Developer and Tester

This team member is responsible for developing the database and testing the quality assurance in the project. Following tasks, which must be achieved:

- Designs and maintains the MariaDB database schema
- Perform continuous testing during each sprint
- Writes and executes unit tests using JUnit

As we are a small team one member may perform multiple roles and share responsibility for code quality, documentation, and continuous improvement.

5.2 Software, Hardware, and Tools

To implement the project successfully, the following software, hardware, and tools are required:

Software Requirements

- Java Development Kit (JDK)
- JavaFX for building the desktop user interface
- JPA for database connectivity
- Git for version control
- JUnit for unit testing
- Jenkins for CI/CD
- Docker for containerization
- Kubernetes for orchestration
- Trello for sprint planning and task management

Hardware Requirements

- Personal computers or laptops

Development Tools

- Integrated Development Environment (IDE) such as IntelliJ IDEA
- MariaDB for database management
- GitHub

5.3 External Resources or Support

The project may require limited external resources or support, including:

- Online documentation and tutorials
- Academic resources and references
- Teacher guidance
- Peer feedback

6. Risk Management

This section identifies potential risks that could affect the successful completion of the project.

6.1 Project requirements may be changed during the project development process.

Description:

During the development process, new ideas or feedback from teachers or students may arise. These changes can lead to adjustments in project requirements, which may affect the project scope and timeline

- Likelihood: Medium
- Impact: Medium
- Mitigation Strategies:
- Regularly review and discuss requirements during sprint planning and sprint review meetings to ensure alignment.

6.2 Database Connectivity Issues

Description:

There may be difficulties in connecting the JavaFX application to the MariaDB database using JPA.

- Likelihood: Low
- Impact: High
- Mitigation Strategies:
- Test the database connection at an early stage of development to identify issues quickly

6.3 Incorrect Grade Calculation Logic

Description:

There is a possibility that mistakes may occur when calculating student averages or applying grading weights. This may affect the result.

- Likelihood: Medium
- Impact: High
- Mitigation Strategies:
- Review calculation logic with other team members to catch errors early
- Check calculations using sample student data and compare them with manual results

6.4 Complete the task within limited time frame

Description:

Team members may sometimes have limited time due to classes, exams, or other responsibilities. These could cause delays in completing sprint tasks or meeting project deadlines

- Likelihood: Medium
- Impact: Medium
- Mitigation Strategies:
 - Set realistic and achievable goals for each sprint based on team availability
 - Regularly monitor progress and task completion using Trello

6.5 Limited Team Experience

Description:

Some team members may not be very familiar with tools that are used in the project at the beginning of the project. This could make certain tasks take longer than expected.

- Likelihood: Medium
- Impact: High
- Mitigation Strategies:
 - Use online tutorials, examples, and official documentation to improve understanding
 - Spend extra time in the early sprints learning and practicing the required technologies

7. Testing and Quality Assurance

This section describes how to make sure that the system works correctly using testing and how to ensure the quality of the project using quality assurance.

7.1 Testing Approach

Testing will be conducted throughout the software development lifecycle. The goal of this testing is to ensure the accuracy of components such as grade calculations, data storage and retrieval, access control and interface.

7.2 Types of testing

Unit testing will be conducted using JUnit to test the individual methods and classes separately. Grade calculation logic, validation of marks, login authentication logic will be tested in this unit testing.

Integration testing will be used to test how different components work together. As an example, this testing method will ensure that the JavaFX interface communicates accurately with MariaDB through JPA. Moreover, it ensures that data is stored, retrieved and displayed correctly.

System testing will be performed to test the complete application. In this testing, the focus is on functional requirements. It is conducted to verify that the application meets all the functional requirements such as the teacher can add, edit and delete student records, the teacher can enter grades and generate report cards and students can only view grades and feedback.

User acceptance testing will be done to validate that the system meets the user expectations. It will be done from the user's perspective and the users, which are teachers and students will test the system.

7.3 Success Criteria

For the system currently to be considered successful, the following conditions must be met:

- All grades and averages are calculated correctly
- No unauthorized access
- No system crashes during use
- Database operations must be completed without data loss
- All JUnit tests pass successfully

7.4 Testing tools and framework

For unit testing, JUnit will be used while for UI testing through manual interaction, JavaFX will be used. Moreover, MariaDB and Git will be used for database testing using sample data and tracking bug fixes and test improvements respectively.

8. Documentation and Reporting

8.1 Documentation

Product vision report is provided to explain the problem statement, target audience, value proposition, vision statement, goals and objectives of the project.

System design documentation will be produced to explain the system architecture, database schema and the UI design.

Additionally, a user-manual (README) will be designed to include separate guidelines for teachers and students with step-by-step instructions along with the screenshots where possible.

Finally, the final project report will be created with a summary of the project overview, development process, challenges and solutions, final outcomes and future improvements.

8.2 Progress reporting

For sprint planning and tracking, Trello will be used to manage backlog, track sprint tasks and to monitor progress. Moreover, regular updates will also be documented in Trello and Sprint review reports.

8.3 Evaluation Criteria

Project plan will be evaluated based on the following criteria:

- Clarity and completeness of the project overview and objectives.
- Realism and detail in the scope, timeline, and resource allocation.
- Thoroughness of risk management and testing strategies.
- Quality and structure of the documentation.
- Timeliness of the submission.